The required amount of rich mixture for starting the engine is given by the starting petrol jet. In the two holes presented in the starter valve one is connected with starter petrol jet and another hole is connected with the starting passage. While starting the engine, for sending rich mixture the bigger size hole in the starter valve is connected to the starting passage. Hence the amount of petrol fuel mixed with the air gets increased and results in formation of rich mixture and enters into the mixing chamber. After the engine is started the starter lever must come to the balanced state. Hence a small hole in the starter valve coupled with the petrol jet is closed and hence the level of petrol fuel is reduced. When the engine comes to its normal operation the startor lever is brought to its off position.

3) **Idling and low speed circuit:** During idling stage the engine will not be accelerated. So the venturi is completely closed by the throttle valve. Hence the vacuum created inside the intake manifold is passed to the idle port located below the throttle valve. Hence the petrol fuel from the idlejet is mixed with air which coming from air jet and finally the mixture is sent to the engine through idle port. Idling and low speed circuit can be seen in Fig 9.5.5(c) Idling and Low Speed Circuit.

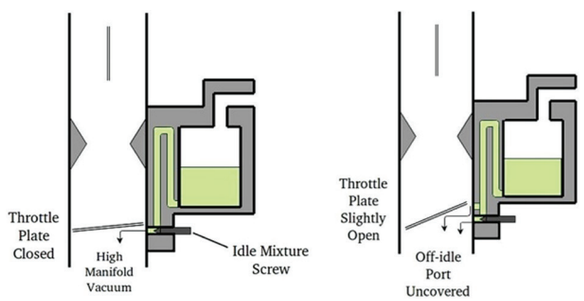



**Fig.9.5.5(c) Idling and Low Speed Circuit**

When driver accelerates the engine for moving the vehicle, the throttle valve gets opened slightly. Because of the above phenomenon vacuum inside the intake manifold is spreaded in to the idle port and slow speed opening. Hence the fuel coming from the petrol jet and air coming from the air jet are inducted to the engine through slow speed opening and idle port.

4) **Normal running circuit:** For increasing the engine speed driver presses the accelerator, at the time throttle valve gets opened further. Hence the required amount of petrol fuel needed for the engine is comes through the main jet and at the same time maximum amount of air is supplied to the venturi. The air is mixed with the petrol fuel and sent to the engine through the throttle valve.

5) **Acceleration circuit:** If the vehicle is to be accelerated the more fuel must be supplied with rich mixture condition for the engine operation. Acceleration pump in the acceleration circuit is used to perform the above process. Acceleration pump is connected to the accelerator pedal with linkages. The acceleration pump which consists of diaphragm coupled with a spring at one side and linkages on other side which is connected to the accelerator pedal. Petrol fuel is supplied from the float chamber to pump chamber with the help of ordinary pump connected in between the both chambers with pump valve.

By giving sudden acceleration to the accelerator pedal the diaphragm inside the pump will acts against its spring tension.



AM_Unit_09.indd 184 31-01-2019 18:02:29